

Logos Experiment 11

Longer Training for Weight Tying + SwiGLU

This note documents the clean rerun of Experiment 11 after restoring the original OpenWebText loader in Cell 4. The experiment trains the best quality-efficient architecture from Experiment 5 for a longer 3000-iteration budget.

Item	Value
Run name	logos_exp11_weight_tying_swiglu_longer_train
Architecture	Decoder-only GPT-style LM with Weight Tying + SwiGLU
Dataset	Original OpenWebText subset, cached as openwebtext_subset_10k.txt
Tokenizer	GPT-2 tokenizer
Training objective	Causal next-token prediction
Training budget	3000 iterations
Main purpose	Confirm whether the best 1500-iteration quality model was undertrained

1. Research Question

The central question for this run was whether the best compact architecture from Experiment 5 would continue improving if trained longer, without increasing parameter count, checkpoint size, or architecture complexity.

Scientific hypothesis: if the Weight Tying + SwiGLU model was undertrained at 1500 iterations, then doubling the training budget to 3000 iterations should reduce validation loss and perplexity while preserving the same model size.

2. Clean Reproducibility Check

Before finalizing this experiment, Cell 4 was corrected back to the original non-filtering OpenWebText loader. The rerun printed:

```
Loading cached original OpenWebText subset...
```

This confirms that the clean Experiment 11 result uses the original OpenWebText subset, not the clean-filtered dataset from Experiment 10.

3. Model Configuration

Component	Value
Model family	Decoder-only causal Transformer
Layers	8
Embedding dimension	256
Attention heads	8
Head dimension	32
Context length	256 tokens
Batch size	64
Tokens per step	16,384

Component	Value
Feed-forward layer	SwiGLU
SwiGLU hidden size	682
Normalization	LayerNorm
Weight tying	Active
Trainable parameters	19,242,400

4. Training Configuration

Setting	Value
Max iterations	3000
Evaluation interval	300
Evaluation iterations	50
Learning rate	3e-4
Weight decay	0.1
Gradient clipping	1.0
Warmup iterations	150
Minimum LR ratio	0.05
Mixed precision	Enabled on CUDA

5. Final Clean Results

Metric	Value
Best iteration	3000
Best validation loss	5.2932
Final validation loss	5.3123
Final validation perplexity	202.82
Checkpoint size	73.45 MB
Peak VRAM	11.18 GB
Training time	280.86 sec = 4.68 min
Generation latency	14.03 ms/token
Generation speed	71.27 tokens/sec
Average GPU power	251.05 W
Estimated GPU energy	19.48 Wh

6. Comparison with Experiment 5

Experiment 5 used the same architecture but trained for only 1500 iterations. Experiment 11 doubled the training budget while keeping the model size and checkpoint size effectively unchanged.

Metric	Exp 5: 1500 iters	Exp 11 clean: 3000 iters	Result
Parameters	19.24M	19.24M	Same
Best val loss	5.7230	5.2932	Improved
Final val loss	5.7247	5.3123	Improved
Perplexity	306.36	202.82	Improved
Checkpoint size	73.44 MB	73.45 MB	Same
Peak VRAM	11.18 GB	11.18 GB	Same
Training time	157.07 sec	280.86 sec	Higher
Generation latency	15.04 ms/token	14.03 ms/token	Improved
Energy	10.49 Wh	19.48 Wh	Higher

7. Quantified Improvements

Comparison	Improvement
Final validation loss vs Exp 5	Reduced by 0.4124
Perplexity vs Exp 5	Reduced by about 33.80%
Generation latency vs Exp 5	Reduced by about 6.72%
Training time vs Exp 5	Increased by about 78.81%
Training energy vs Exp 5	Increased by about 85.70%
Parameter count vs original baseline	About 40.17% fewer parameters
Checkpoint size vs original baseline	About 40.14% smaller checkpoint
Perplexity vs original baseline	About 36.30% lower perplexity

8. Interpretation

Experiment 11 shows that the Weight Tying + SwiGLU model was undertrained at 1500 iterations. When trained for 3000 iterations, it achieved a major quality improvement while preserving the compact 19.24M-parameter size and 73.45 MB checkpoint.

The most important result is that model quality improved substantially without increasing model capacity. This supports the Logos project goal of improving capability per parameter and capability per storage byte.

The cost is higher training time and energy. This is expected because the training budget was doubled. However, inference-related model size stayed fixed, and generation latency slightly improved compared with Experiment 5.

9. Scientific Conclusion

Experiment 11 is the best quality result so far. It establishes that longer training of the compact Weight Tying + SwiGLU architecture significantly improves validation loss and perplexity without increasing parameter count or checkpoint size.

Clean conclusion: the best current Logos quality model is a 19.24M-parameter decoder-only Transformer using weight tying and SwiGLU, trained for 3000 iterations on the original OpenWebText subset.

10. Current Best Model Status

Category	Current best
Best quality model	Experiment 11: Weight Tying + SwiGLU, 3000 iterations
Best final validation loss	5.3123
Best perplexity	202.82
Compact checkpoint	73.45 MB
Parameter count	19.24M

11. Next Step

The next scientific step should be a matching longer-training confirmation run for Experiment 9, the best efficiency-focused architecture. This will determine whether the wider-shallower efficient model also benefits strongly from 3000 iterations.

Recommended next experiment: Experiment 12 - Wider-Shallower + Weight Tying + SwiGLU, 3000 iterations.